

CRM AI Lead Assistant

Specification

Odoo module: `crm_ai_lead` | 2026-08-24

1. What it is

A chat window inside Odoo CRM. You type a message describing one or more clients, and it creates, deletes, or updates the matching `crm.lead` records. No confirmation step — it acts right away.

2. How it's built

Two folders:

- `crm_ai_lead/` — the Odoo module (models, views, chat widget)
- `AI_operations_system/` — the AI logic (`nodes.py`), imported by the module

The AI logic calls a local LLM (Ollama, model `qwen2.5:7b`) running on the same machine at `localhost:11434`.

Leads are written to Odoo two ways:

- From inside Odoo** (the chat widget): direct ORM call, e.g. `env["crm.lead"].create(...)`
- From outside Odoo** (the test script, `process.py`): XML-RPC, like any external app

Why two ways: calling XML-RPC back into the same Odoo process that's already running caused a bug — the new lead was created in a separate transaction the current one couldn't see yet, so linking it back failed. Using the direct ORM call when already inside Odoo avoids that.

3. Data model

Model	Fields	Purpose
<code>crm.ai.lead.wizard</code>	<code>message_ids</code> , <code>state_json</code> , <code>lead_id</code>	One chat session. Deleted and started fresh every time you open "New Lead AI".
<code>crm.ai.lead.wizard.message</code>	<code>wizard_id</code> , <code>role</code> , <code>content</code>	One chat message (user or assistant).

A created lead fills these Odoo fields: `name`, `contact_name`, `email_from`, `phone`, `description`, `tag_ids`, `priority`, `partner_name`, address fields, `website`, `country_id`.

Tags matter: the operation (buy/sell/rent) and the property type (villa/house/apartment) are both saved as real Odoo tags, not just text. That's what lets delete and update find the right leads later.

4. Fields you can give it

Field	Required?	Values
name	Yes — only this one	any text
tags	No	buy / sell / rent
property_type	No	villa / house / apartment
priority	No	0-3 (shown as stars)
email, phone	No	—
company_name, street, street2, city, country, website	No	—
notes	No	anything else worth keeping (budget, timeline, etc.)

If a name is missing for a mentioned client, it's asked for separately — other clients in the same message with a name are still created right away.

5. The three operations

Operation	What it needs	Example
Create	Any of the fields above, per client	"Add Sarah, wants to rent an apartment"
Delete	A name and/or tags/property type to match on	"Delete whoever wants to sell a house"
Update	Same matching, plus what to change	"Change Ahmed's phone to 0612345678"

Important rule: the AI never guesses which leads match by remembering the conversation. It only pulls out the criteria you stated (name / tags / property type), and a real database search finds the matching leads. This was added after the AI once guessed wrong and deleted the wrong lead.

6. Chat widget

One component handles everything: message list, text box, and send button together. When you hit send, your message shows immediately and a "typing" animation plays while it waits for the real reply — instead of the screen doing nothing for several seconds.

7. Known limitations

- No "are you sure?" step before delete or update
- Leads made before tags were added for property type only have the operation tag

- Name matching is "contains", so similar names can overlap
- The AI model can still make mistakes — this design limits the damage, it doesn't remove the risk
- Editing Odoo views needs a module upgrade; editing Python code needs a server restart — one doesn't cover the other

8. Files

File	Role
crm_ai_lead/models/crm_ai_lead_wizard.py	Chat session logic
crm_ai_lead/views/crm_ai_lead_wizard_views.xml	Chat page layout
crm_ai_lead/views/crm_lead_views.xml	"New Lead AI" button on Leads list
crm_ai_lead/static/src/wizard/chat.js/.xml/.css	Chat widget (front end)
AI operations system/nodes.py	AI logic — reading messages, create/delete/update
AI operations system/model.py	Talks to Ollama
AI operations system/odoo_client.py	Talks to Odoo over XML-RPC (outside Odoo only)
AI operations system/process.py	Terminal version, for testing without the web page